Basic Mathematics

Lecture 0: Introduction

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What is this course about?

structure **space** **change**

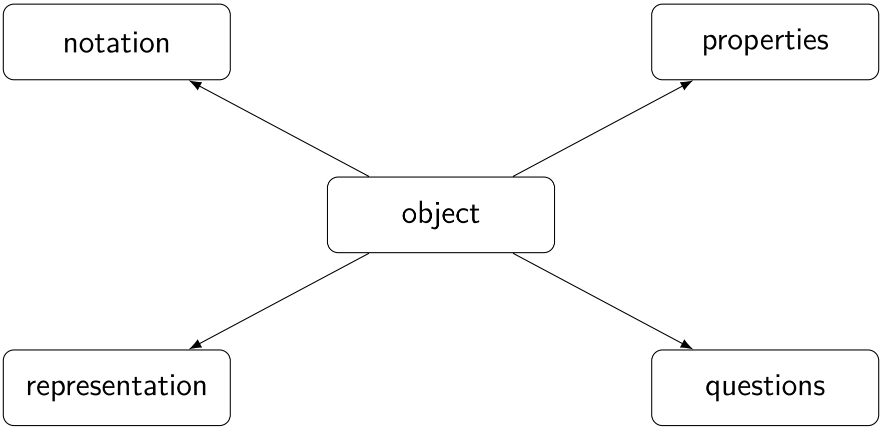
Algebra → Geometry → Calculus

Not only formulas

$$\begin{array}{ccc} \text{formula} & \text{rule} & \text{answer} \\ \neq & \neq & \neq \\ \text{idea} & \text{reason} & \text{interpretation} \end{array}$$

Mathematics starts when we ask: **why?**

A mathematical object



First identify the object. Then choose the method.

Three languages

Linear algebra

$$Ax = b$$

vectors, matrices, systems

Analytic geometry

$$ax + by + c = 0$$

lines, planes, distance

Calculus

$$f'(x), \qquad \int_a^b f(x)\,dx$$

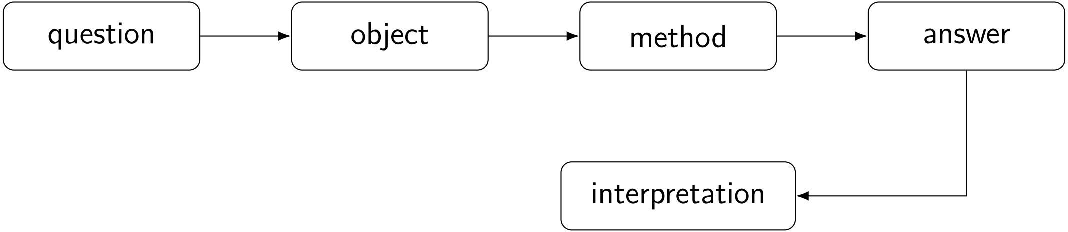
change and accumulation

What should we ask?

- What is known?
- What is unknown?
- What is fixed?
- What can vary?

- What is the object?
- What is the structure?
- Which step is allowed?
- What does the result mean?

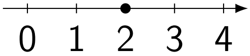
A problem is not only an answer



One idea, many forms

algebraic $\longleftrightarrow$ geometric $\longleftrightarrow$ conceptual

$2x + 3 = 7$



balance

transformation

How to use the PDF

Slides

map of the lecture
ideas, formulas, examples

Notes

full explanations
definitions, details, reasoning

Learn from the notes. Use slides to follow the lecture.

Course map

- Matrices
- Determinants
- Matrix inversion
- Systems of linear equations
- Vectors and coordinates
- Lines and planes

- Sequences and functions
- Limits and continuity
- Derivatives
- Integrals
- Applications of calculus
- Problem sets

Main habit

Ask good questions.

Then compute.